



Characteristics and influencing factors of loess terraces' preferential flow under different typical vegetation cover

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Abstract

Background and Aims Terraces on the Loess Plateau perform vital ecological roles in soil and water conservation. However, different vegetation types have created soil fissures and large pores, affecting groundwater infiltration dynamics and preferential flow characteristics. This study examines preferential flow patterns across four vegetation types in the

Loess Plateau's terraced areas, aiming to identify the main factors influencing these patterns.

Methods The research utilized a bright blue dye as a tracer and image processing technology to analyze soil preferential flow under four different vegetation types. It also quantitatively assessed the relationship between environmental factors and preferential flow characteristics.

Results Substrate flow depths were shallower in caragana fields and woodlands than in grasslands and potato lands. Preferential flows in caragana fields and woodlands were more differentiated, displaying tubular and patchy patterns, predominantly dominated by vertical percolation. In contrast, grasslands and potato lands mostly had lateral infiltration, with dendritic and clustered flow patterns. The depth of preferential flow varied with vegetation type: 20–50 cm in grasslands, 20–40 cm in woodlands, 20–60 cm in caragana fields, and 20–60 cm in potato lands. Maximum staining depths were 55.1 cm, 38.5 cm, 54.0 cm, and 56.2 cm, respectively. The spatial distribution of stained areas and flow paths was influenced by the distribution of plant root systems and soil properties, with soil porosity identified as the primary determinant of preferential flow.

Conclusions Soil properties and plant, root distribution, influenced by vegetation type, are crucial in shaping preferential flow characteristics on the Loess Plateau.

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Introduction

Terracing represents an essential technique for soil and water conservation, primarily employed on the sloping terrains of mountains and hills. This method involves the creation of horizontal spaces that are specifically designed for cultivation or planting activities (Chen et al. 2016). The width of these terraces is carefully adjusted according to the slope's gradient, rendering it a prevalent practice in regions such as the Loess Plateau (Wei et al. 2021) and a distinct manifestation of human intervention across global landscapes (Tarolli et al. 2014; Wei et al. 2016). By modifying agricultural practices and crop selections, terracing significantly improves the physical properties of soil (Qiang et al. 2020; Xiao et al. 2018), enhances the activities of root systems and soil fauna (Xiao et al. 2018), and encourages the development of diverse vertical preferential flow pore structures (Ran et al. 2020). The arrangement of fissures and pores within terraced areas plays a critical role in the transport of soil water (Mesfin et al. 2018). In pursuit of advancing modern agriculture, ecological conservation, and fostering high-quality regional growth (Zhang et al. 2017), an increasing number of researchers have begun to investigate the geomorphology of terraces, soil resources, and terrace farming techniques (Fernandes and Gontijo 2020; Jiao et al. 2020; Wu et al. 2023; Zhang et al. 2014). However, a significant research gap exists concerning the influence of typical vegetation cover on the soil structure and water transport within terraced fields.

Preferential flow is a commonly observed phenomenon in terrace soil, markedly affecting the water cycle, soil erosion, and the risk of mountain-related disasters. In the field of ecohydrology, preferential flow is considered pivotal (Jiang et al. 2020). Preferential flow phenomenon epitomizes the transition of water and solute transport from homogeneous to non-homogeneous states, representing a non-equilibrium flow process. During this process, water and solutes circumvent the soil matrix, infiltrating deeper soil layers or reaching the groundwater more rapidly (Dekker et al. 1999). As a result, the rate of water flow

and solute transport within preferential flow channels significantly exceeds that within the matrix zone (Hendrickx and Flury 2001), profoundly affecting the recharge of groundwater by rainfall, the formation of runoff, soil water infiltration, and water redistribution (Liu et al. 2016, 2014). Various factors influence preferential flow, including soil texture, structure, porosity, water repellency, bulk density, organic matter content, vegetative roots, presence of gravel, solute application rates, irrigation practices, tillage methods, and dry–wet–freeze–thaw cycles (Hu et al. 2023; Zhang et al. 2016). Despite its importance, research focusing on the impacts of roots and soil on preferential flow remains limited (Bundt et al. 2001). Existing studies have primarily analyzed the correlation between preferential flow and root density, with most research concentrated on forest and agricultural lands (Fuhrmann et al. 2019; Guan et al. 2023), leaving a notable gap concerning loess terraces.

Preferential flow plays a critical role in the rapid distribution of water through the soil profile, facilitating its extension to deeper horizons (Alaoui et al. 2011; Tang et al. 2020). The specific geometry of preferential flow significantly determines which segments of the soil profile interact with the infiltrating water. The exchange of water between macropores and the soil matrix is crucial for the accumulation and retention of water within the soil profile. This process is governed by the connectivity and interfacial characteristics of these regions (Weiler and Flühler 2004; Weiler and Naef 2003). The Brilliant Blue dye tracer method serves as a tool to visualize the preferential flow paths of infiltrating water and to evaluate the connectivity within porous systems (Jačka et al. 2018; Zhu et al. 2019). Nonetheless, this the method has limitations, such as potential adsorption issues and diminished mobility in smaller pores when compared to untreated water (Nobles et al. 2010; Wang and Zhang 2011). While the dye tracer method does not provide quantitative data on water flow, it offers invaluable qualitative insights into the characteristics of preferential water flow.

The direct measurement of preferential flow is inherently challenging due to the heterogeneous and dynamic nature of soil water movement. Techniques involving stained tracers, such as the use of Brilliant Blue dye, are frequently employed to delineate preferential flow patterns. Despite this, systematic evaluations of preferential flow and its influencing

factors across different types of vegetation in terraced landscapes, especially on the western Loess Plateau, remain sparse. This study employs dye tracing, high-resolution image analysis, and assessments of soil physicochemical properties and root distributions to quantify and contrast preferential flow under four typical vegetation covers—grassland, woodland, caragana fields, and potato land. In contrast to earlier research that focused on individual vegetation types, this investigation underscores the influence of vegetation structure on the dynamics of preferential flow. It hypothesizes that the type of vegetation dictates the patterns of preferential flow in terraced soils through variations in root architecture and soil properties. The findings of this study offer novel insights into the mechanisms of water movement in terraced loess soils and provide guidance for soil and water conservation strategies in semi-arid regions.

Materials and methods

Study area

The study area is situated within the Guanchuan River basin located in the western segment of the Loess Plateau, and specifically within the Anjiagou sub-watershed comprehensive observatory at Dingxi City, Gansu Province (coordinates: 35°33'02"N to 35°35'29"N, 104°38'13"E to 104°40'25"E) (Fig. 1) (Yang et al. 2022). This location is emblematic of the Loess Hills and Gullies landscape unit. The elevation of this area ranges from 1900 to 2250 m. The predominant soil type is loess, where belongs to the area's mid-temperate semi-arid climate which has an average annual temperature of 6.3 °C and an average annual precipitation of 427 mm. Precipitation is predominantly concentrated during the flood season, accounting for over 60% of the annual total and contributing to significant inter-annual variability. The

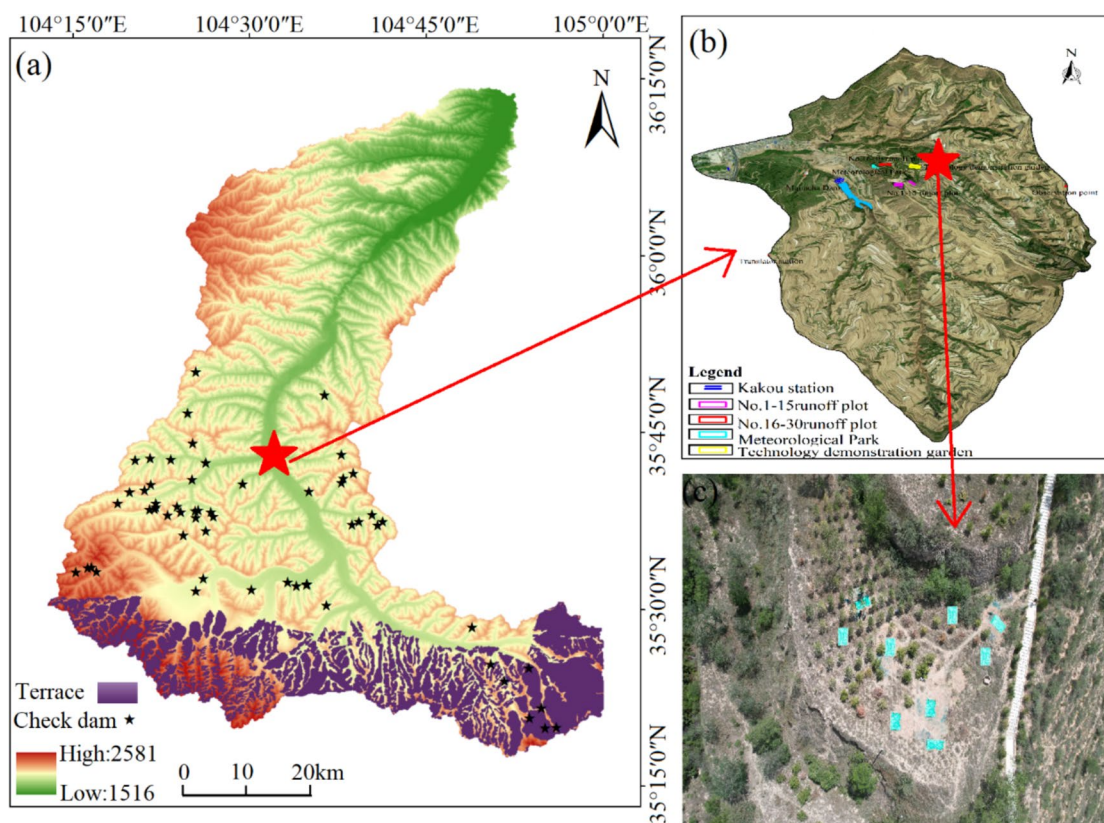


Fig. 1 Location map of the study area: (a) distribution of terraces in the Guanchuan River basin; (b) map of the Anjiagou sub-watershed; (c) map of the test site

annual evaporation rate is approximately 1510 mm. The area receives an average of 2408.6 h of sunshine annually and enjoys a frost-free period of 141 days. The Anjiagou sub-watershed features seasonal ditches that are primarily recharged by precipitation, with runoff volume heavily dependent on precipitation levels. The dominant vegetation is temperate grassland. The land use includes cultivated land, forest, grassland, and other types. Soil depth under the vegetation typically averages about 50 cm, primarily composed of loess. Soil erosion in the area is predominantly caused by hydraulic action, although gravitational erosion is also present.

Experimental design

This study focused on four typical vegetation types found on representative terraces within the Anjiagou sub-watershed of the Guanchuan River Basin on the Loess Plateau: grassland, woodland, caragana field, and potato land. These vegetation types were selected as research subjects. A field survey and site selection were conducted in September 2023 by the research team to ensure that all experimental plots shared similar slope orientations and sunlight exposure. For each vegetation type, a 10 m × 10 m experimental plot was established. Staining tracer experiments were carried out during periods devoid of rain, with at least a one-day interval before and after each experiment to preclude interference from natural precipitation on the infiltration process. For each vegetation type, three replicate staining tracer experiments were conducted. Each replicate comprised five vertical staining profiles, the number and depth of which were determined based on the actual infiltration depth of the dye. The staining tracer experiment involved several: preparing a double-ring infiltration pit, preparing and applying the dye tracer solution, covering the pit, excavating the stained profile, photographing and sampling the soil, and ultimately backfilling the pit.

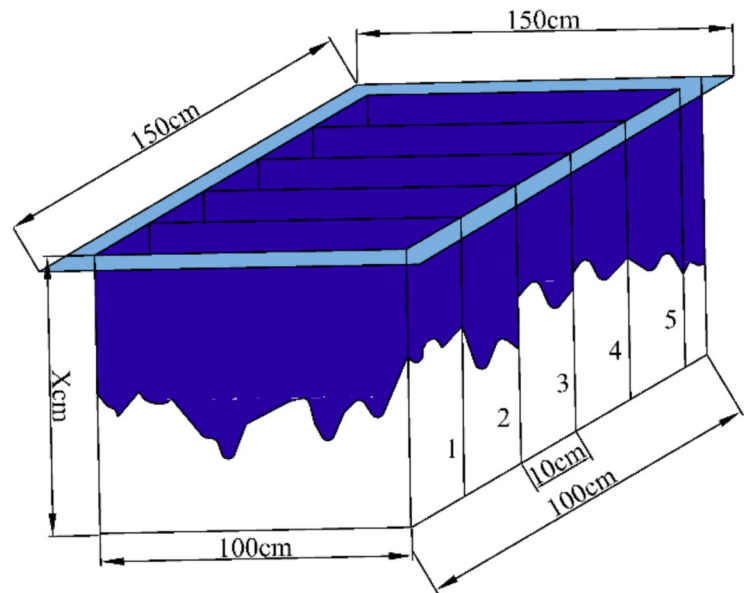
Dye tracer and imaging process

The Brilliant Blue solution, with a concentration of 6 g/L, was selected as the dye tracer owing to its excellent visibility, mobility, and minimal toxicity (Liu et al. 2018; Yi et al. 2020). Prior to the experiment, surface weeds were removed and a 20 cm layer of topsoil was excavated from the designated test

site. A test pit was subsequently prepared, featuring dimensions of 1.5 m × 1.5 m for the outer boundary and 1 m × 1 m for the inner boundary. The inner boundary received a uniform application of 40 L of the Brilliant Blue solution. In contrast, the outer boundary was sprayed with an equivalent volume of clean water, ensuring that the infiltration rates of the clean water matched those of the Brilliant Blue solution. Throughout the application of the dye, no discernible water accumulation was observed on the surface of the soil within either the inner or outer boundaries (Jiang et al. 2017). Following the application, the site was covered with a waterproof cloth to mitigate the effects of surface evaporation and potential erosion due to rainfall, thereby preserving the integrity of the experiment (Jiang et al. 2015). After a 24-h infiltration period, the cloth was removed, and a working platform was constructed along one side of the test pit to aid in the excavation of soil sections (Jiang et al. 2020; Wang et al. 2020).

Vertical soil staining profiles were methodically excavated as depicted in Fig. 2, maintaining a lateral spacing of 10 cm between each profile. Each sampling plot comprised five staining profiles, each 1 m in width. To accommodate spatial variability in infiltration depth, the maximum excavation depth for each profile was established based on the clearly visible lower boundary of Brilliant Blue dye penetration. To ensure consistency across different vegetation types and replicates, identical infiltration durations, dye application volumes, and observation times were meticulously adhered to across all plots. The depth of dye penetration for each profile was meticulously recorded. Subsequent image analysis and quantification were standardized by normalizing to the profile depth, facilitating consistent comparisons across profiles. The soil profiles were captured using a digital camera (Canon, EOS 80D, 6000 × 4000 pixels) equipped with a sunshade and a colorimetric card. Geometric corrections were applied to minimize noise and enhance the quality of the digital images. This study documented the stained vertical sections using digital photography. All images of the vertical soil profile were processed using Adobe Photoshop software 2018 (available from <https://www.photoshop.com/en>) to adjust hue, saturation, and grayscale levels. The red–green–blue (RGB) images were subsequently

Fig. 2 Schematic diagram of soil dye surface excavation



converted to grayscale and then to black-and-white binary images for further analysis. These binary images were used to create two-phase diagrams, which formed the basis for further image vectorization and quantitative analysis. The binarized dyed images were imported into Image-Pro Plus 6.0 software for morphological recognition and data extraction. The extent of staining within the soil profile was quantified by analyzing the grayscale values of individual pixels. By correlating the pixel count with the actual area represented by each pixel, the stained area at various depths of the profile and the corresponding staining ratio were computed. This staining ratio served as an indicator of the proportion of dye within the vertical soil profile, with a higher stained area ratio suggesting more extensive water infiltration. Image analysis was conducted at 1 mm depth intervals. The stained area ratio was calculated by comparing the number of dyed pixels to the total area of the profile, with each pixel representing an area of 1 mm². Zones of prominent staining were identified as areas of preferential flow, characterized by the rapid and uneven movement of water and solutes through macropores and root channels. The identification of such pathways was based on the visual distinction of staining patterns, in accordance with established methodologies (Aeby et al. 1997; Flury et al. 1994).

Soil sampling and analysis

Soil samples were collected from soil horizons at depths of 0, 10, 20, 30, 40, and 50 cm after removing a 30-cm-deep overburden in each typical vegetation sample plot. These samples were analyzed to measure various properties such as soil bulk density (BD , g cm⁻³), soil porosity (SP , %), initial water content, soil saturated water content ($SSWC$, %), saturated hydraulic conductivity (K_s , mm h⁻¹), plant root length density (RLD , mm cm⁻³), and root surface area density ($RSAD$, mm² cm⁻³). Each stratum had three replicates (Reynolds and Elrick 1990). The assessments of BD , SP , $SSWC$, K_s , RLD , and $RSAD$ were conducted using a ring knife with an inner diameter of 50.46 mm, a height of 50 mm, and a volume of 100 cm³, following the methodology described by Chen (2005). The K_s was determined using the constant head method, while the initial water content was measured through a drying method. The weight of the empty ring knife (W_0 , g) was recorded. In the laboratory, the samples were saturated with distilled water, and the weight of the saturated soil (W_1 , g) was measured. After drying the samples at 105 °C, the weight of the ring knife containing the dry soil (W_2 , g) was recorded. All root-soil mixture samples underwent separation by elutriation in the laboratory. Plant roots were categorized into six diameter classes: 0 to 0.5 mm, 0.5 to 1 mm, 1 to 2 mm, 2 to 3 mm, 3 to 4 mm, and greater

than 4 mm (Smit et al. 1999). The root length (L_{root} , mm) was measured using an electronic measurement system, and the length of fine (less than 2 mm) was obtained with a root scanner. The root surface area (S_{root} , mm²) was then calculated based on these measurements.

$$BD(gcm^{-3}) = (W_2(g) - W_0(g))/100(cm^3) \quad (1)$$

$$SP(\%) = (1 - BD(gcm^{-3})/Particle\ density(2.65gcm^{-3})) \times 100\% \quad (2)$$

$$SSWC(\%) = (W_1(g) - W_2(g))/(W_2(g) - W_0(g)) \times 100\% \quad (3)$$

$$RLD(mm\ cm^{-3}) = L_{root}(mm)/100(cm^3) \quad (4)$$

$$RSAD(mm^2cm^{-3}) = S_{root}(mm^2)/100(cm^3) \quad (5)$$

Evaluation indices for preferential flow

- (i) Dye-stained coverage (D_c , %) reflects the proportion of the stained area in the profile image relative to the total area, indicating the extent of seepage. Generally, an increase in the degree of preferential flow correlates with a decrease in the stained area ratio (Flury et al. 1994). For this study, each calculation unit was defined with a depth of 1 cm and a width of 100 cm. The calculation formula is as follows:

$$Dc = D_{pf}/S \times 100\% \quad (6)$$

where D_c is the dye-stained coverage (%); S is the total area of the profile image (cm²); and D_{pf} is the area of the stained portion of the profile image (cm²).

- (ii) The uniform percolation depth (U_{niFr}) measures the depth to which matrix flow penetrates (Tobella et al. 2014; Van Schaik 2009). It is determined when $D_c \geq 80\%$ is observed in the profile. Matrix infiltration depth indicates that infiltration is primarily governed by matrix flow before reaching that depth. Consequently, A higher degree of preferential flow, typically results in a shallower matrix infiltration depth.
- (iii) Preferential flow fraction (P_{Fr}) represents the proportion of preferential flow within the total seep-

age. As preferential flow increases, so does this fraction (Van Schaik 2009). The calculation formula is:

$$P_{Fr} = (1 - 100U_{niFr}/S_{Tot}) \times 100\% \quad (7)$$

where P_{Fr} is the preferential flow fraction (%); S_{Tot} is the total dye-stained area (cm²); U_{niFr} is the uniform infiltration depth (cm); 100 is the width of

the profile (cm); and $100U_{niFr}$ is the area occupied by substrate flow.

- (iv) The Length Index (L_i) accounts for the cumulative absolute differences between adjacent D_c values along the soil profile. A higher index value suggests a greater degree of heterogeneity in the seepage pattern, which correlates with increased preferential flow (Tobella et al. 2014). It is calculated using:

$$L_i = \sum_{i=1}^n |D_{C(i+1)} - D_C| \quad (8)$$

where L_i is the length index (%); D_{ci} is the dye-stained coverage of layer i in the profile (%); $D_{C(i+1)}$ is the dye-stained coverage of the subsequent layer ($i+1$) in the profile (%).

Utilizing the aforementioned indices, it was observed that both the stained area ratio and matrix infiltration depth exhibited negative correlations with the degree of preferential flow. Conversely, the preferential flow fraction, flow path number index, and coefficient of variation demonstrated positive correlations. A rigorous evaluation of these indices enables the construction of an integrated index to accurately reflect the intensity of preferential flow. This comprehensive index is derived by standardizing each indicator using the range method, calculating the variance of each standardized indicator, determining their corresponding weights, and ultimately integrating these into a holistic evaluation metric. The detailed methodology and computational steps are outlined below.

- (v) The coefficient of variation C_v quantifies the variability in soil staining across different depths of a soil profile. It is indicative of the profile's inho-

mogeneity in dye distribution, meaning a higher degree of preferential flow results in a greater coefficient of variation. The formula for calculating C_v is outlined in Eq. (9),

$$C_v = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (D_{Ci} - \bar{D}_C)^2 / \frac{1}{n} \sum_{i=1}^n D_{Ci}} \quad (9)$$

where C_v is the coefficient of variation; $\bar{D}_C$ mean dye-stained coverage (%) of stained profiles.

In the analysis of five selected indicators of preferential flow, it was noted that dye-stained coverage and the uniform percolation depth are inversely related to the degree of preferential. Conversely, the preferential flow fraction, length index, and coefficient of variation all correlate positively with it. By thoroughly examining these indicators, one can develop an integrated index to assess the overall intensity of preferential flow. This process involves standardizing each indicator using the range method, calculating the variance of each standardized indicator, assigning appropriate weights, and ultimately combining them into a comprehensive evaluation index for preferential flow. The detailed methodology is as follows:

(1) Data Standardization

For indicators that show a positive correlation:

$$Z_{ij} = (X_{ij} - X_{min}) / (X_{max} - X_{min})$$

For indicators that show a negative correlation:

$$Z_{ij} = (X_{max} - X_{ij}) / (X_{max} - X_{min}) \quad (10)$$

where Z_{ij} represents the standardized value of each indicator, X_{ij} denotes the raw data of each indicator, X_{min} and X_{max} are the minimum and maximum values within the indicator data respectively, i indicates the i th value in a particular indicator data ($i = 1, 2, \dots, m$), and j represents the number of indicators ($j = 1, 2, \dots, 5$).

(2) Square Error Calculation

The mean value of each index data $E(G_j)$:

$$E(G_j) = 1/m \sum_{i=1}^m Z_{ij} \quad (11)$$

The mean square error of each index data $\sigma(G_j)$:

$$\sigma(G_j) = \sqrt{1/m \sum_{i=1}^m (Z_{ij} - E(G_j))^2} \quad (12)$$

(3) Weight Coefficient

The weight coefficient for each index data W_j :

$$W_j = \sigma(G_j) / \sum_{j=1}^5 \sigma(G_j) \quad (13)$$

(4) Calculation of the Preferential Flow Evaluation Index (PFI).

For each stained profile, the *PFI* is calculated as follows:

$$PFI = 1/m \sum_{j=1}^n \sum_{i=1}^m Z_{ij} W_j \quad (14)$$

Statistical analysis

In this study, the experimental data were initially compiled using Excel 2019. Binary images, processed in Adobe Photoshop, were analyzed with Image-Pro Plus to derive bitmap-based quantitative data of stained regions along soil profiles. The statistical analyses were conducted using SPSS 27.0. Prior to performing a one-way analysis of variance (ANOVA) to examine differences in soil properties and root parameters among various vegetation types, tests were carried out to verify the assumptions of normality and homogeneity of variances. For variables that satisfied these assumptions, ANOVA was executed, followed by post-hoc tests using the least significant difference method to ascertain differences among groups at a significance level of $P < 0.05$. Each vegetation type consisted of three replicate plots, each treated as an independent observational unit. Variability within and between profiles was analyzed to address spatial heterogeneity in root distribution and infiltration patterns. Origin 2023 software facilitated linear fitting, correlation analysis, and graphical representation of the data.

Results

Characteristics of physical and hydraulic parameters

In the study area, within the terraced loess layer (0–60 cm), variations in the physical and hydraulic properties of the soil at different depths were observed across various vegetation types as illustrated in Figs. 3 and 4. The analysis of soil profiles with more active root systems typically revealed a lower *BD*, and correspondingly, higher values of *SSWC* and *SP*, suggesting a potential relationship between root distribution and soil loosening. However, this pattern was not uniformly consistent across all vegetation types. Among the four distinguished vegetation types, grassland soils displayed the highest variability in physical and hydraulic properties, with coefficients

of variation reaching up to 3.8%, 3.6%, 2.7%, and 1.8% for *BD*, *Ks*, *SSWC*, and *SP*, respectively. In general, soil *BD* was observed to be higher in areas under grassland and potato cultivation, whereas *SP* and *SSWC* were elevated in areas under woodland and caragana fields. The caragana field demonstrated the least variability among all measured properties, indicating relatively stable soil conditions. These findings underscore that vegetation type significantly impacts soil properties, although no clear linear trend was not evident across all parameters.

Differences in vegetation root characteristics

In arid and semi-arid landscapes, the pore channels created by both living and deceased plant roots serve as essential conduits for the preferential flow of water

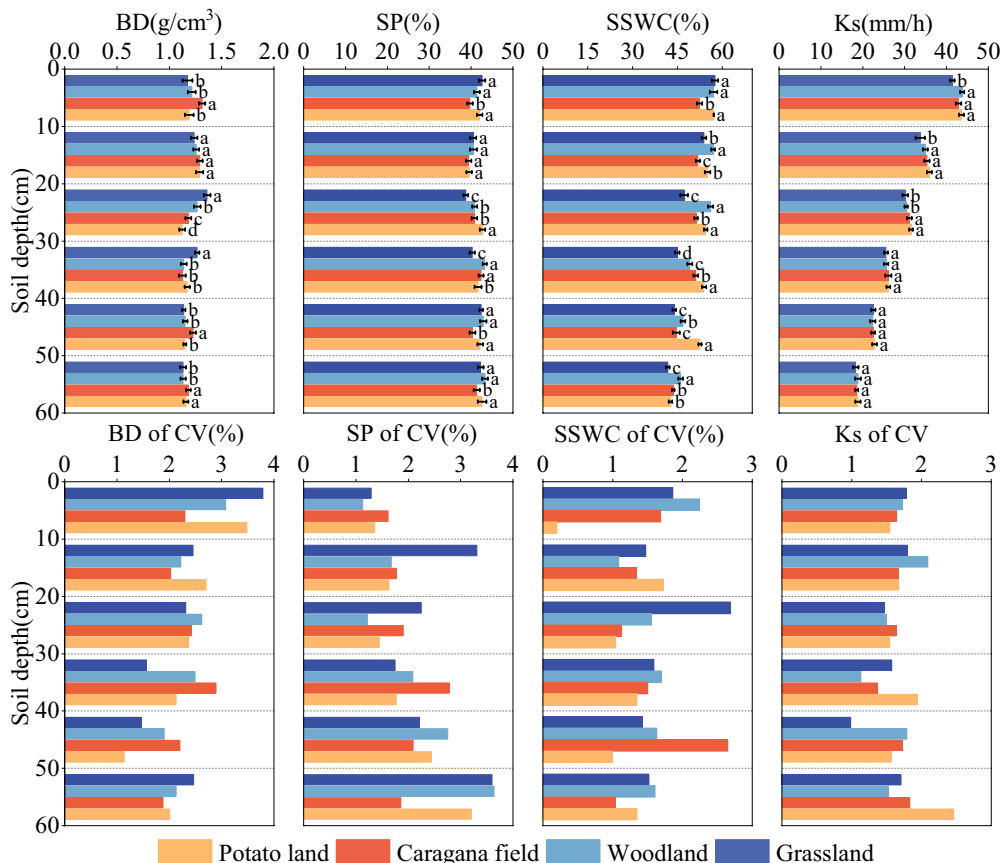
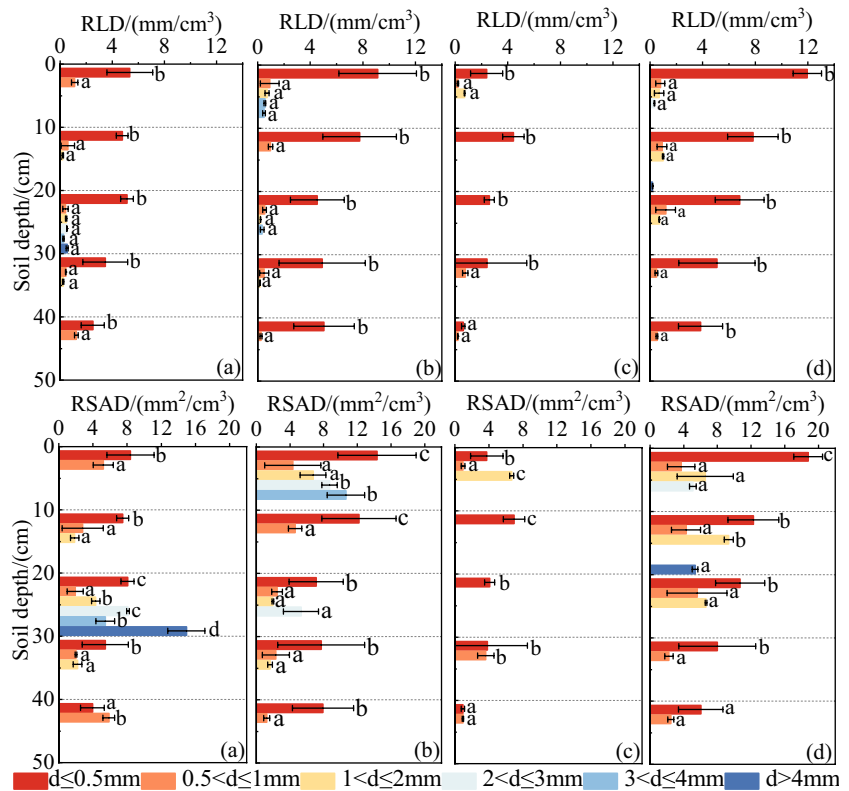


Fig. 3 Changes in bulk density (*BD*, g cm^{-3}), soil porosity (*SP*, %), saturated soil water content (*SSWC*, %), soil saturated hydraulic conductivity (*Ks*, mm h^{-1}), and coefficient of variation (*Cv*, %) in the 0–60 cm soil layer under four typical veg-

etation covers (Error bars represent the standard deviation of the mean. Different lowercase letters indicate significant differences ($P < 0.05$) between samples within the same soil layer)

Fig. 4 Characteristics of *RLD* and *RSAD* in the 0–50 cm soil layer under typical vegetation coverages. (a) woodland; (b) caragana field; (c) potato land; (d) grassland (Error bars represent the standard deviation of the mean values. Distinct lowercase letters denote significant differences among root diameters within the same soil layer ($P < 0.05$))



and solutes within the soil (Mair et al. 2022; Zhang et al. 2022b, 2018). *RLD* is a crucial metric that reflects the growth status of a plant's subterranean portion, while *RSAD* indicates the extent and distribution of a plant's underground root system. Figure 4 delineates the distribution of *RLD* and *RSAD* across six root diameter categories (0–0.5 mm, 0.5–1 mm, 1–2 mm, 2–3 mm, 3–4 mm, and > 4 mm) within the 0 to 50 cm of soil under four distinct types of vegetation: woodland, caragana field, potato land and grassland.

Collectively, the aggregate *RLD* of plants subjected to different land uses exhibited a hierarchy of grassland > caragana field > woodland > potato land, whereas *RSAD* rankings manifested as caragana field > grassland > woodland > potato land. These trends suggest that grassland vegetation fosters a higher *RLD*, whereas caragana fields enhance *RSAD*, both contributing to the development of preferential flow pathways from structural and functional standpoints. Predominantly, roots with diameters ranging from 0 to 0.5 mm were most prevalent in all four vegetation types, particularly in the soil layer extending

from 0~to 30 cm, where they accounted for over 70% of the total *RLD* and *RSAD*. This layer also coincided with the highest concentration of preferential flow paths as identified in staining tracer tests, underscoring the pivotal role of fine roots in the initial stages of water infiltration through the creation of numerous micropores. Conversely, the structural stability, connectivity, and durability of microporous channels constructed by coarse roots significantly influence the formation of preferential flow. The vertical stratification of root diameters further unveils the dual strategies employed by plant roots to regulate water transport: superficial fine roots expedite the rapid infiltration of precipitation through the formation of dense, transient small-scale networks, whereas deeper coarse roots ensure sustained water conduction by constructing a small amount of establishing fewer, but hydraulically stable, vertical macropores. This hierarchical root structure epitomizes a strategic adaptation for optimizing soil water movement in terraced arid and semi-arid environments. In conclusion, the diameter distribution characteristics of *RLD* and *RSAD* not only elucidate the pivotal role of plant roots

in establishing preferential flow channels, but also demonstrate how different vegetation types collaboratively refine soil hydrological processes through a tiered amalgamation of root morphologies. This insight is crucial for comprehending the mechanisms of water transport in terraced soils of arid regions.

Morphological characteristics of soil dyeing

Dye-stained coverage

This research utilized the binary image of preferential flow features from processed vertical stained profiles

as the foundational map. The proportion of stained area across various typical vegetation profiles was charged with respect to soil depth (Fig. 5). The findings reveal substantial variations in the characteristics of preferential flow in soils at four typical vegetation sampling locations. In the samples, the preferential flow paths within the soils of caragana fields and woodlands were observed to develop with increasing soil depth, displaying pronounced differentiation characteristics. These preferential pathways predominantly manifest in tubular and patchy forms, primarily extending longitudinally. Among the studied sites, grasslands ranked second in terms of differentiation,

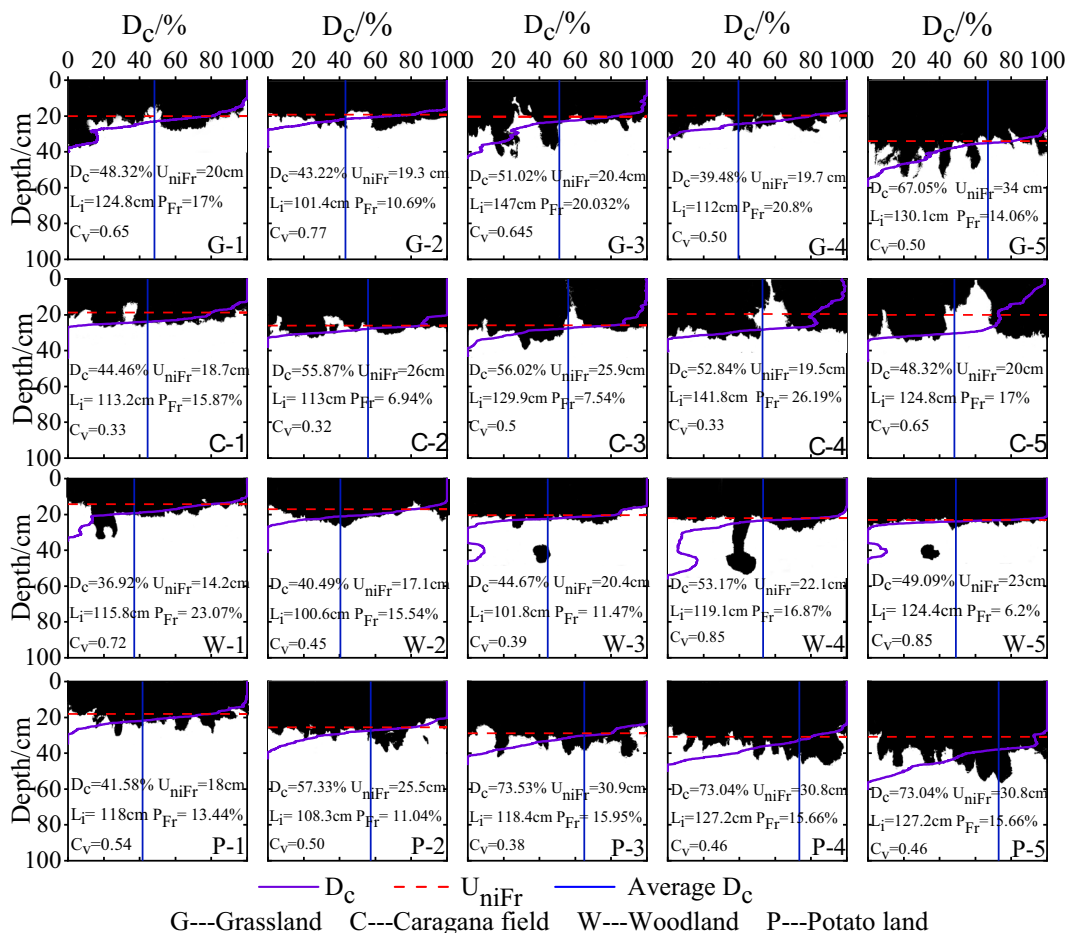


Fig. 5 Analysis of vertical staining and dye-stained coverage across different vegetation types (Variation in dye-stained area ratio along depth at 1 mm intervals. The curve represents the depth-related change in stained coverage. A horizontal dashed line marks the matrix infiltration depth, defined as the depth at which the stained area ratio reaches 80%; in instances of mul-

tiple intersections, the deepest point is considered the infiltration depth. A vertical solid line indicates the average stained coverage throughout the entire profile, and the number of intersections with the stained coverage curve reflects the number of peaks in preferential flow.)

while potato lands exhibited the least differentiation. The preferential flow of soil water during infiltration was more pronounced in vegetation used for soil and water conservation, and the extent of morphological differentiation was greater.

Figure 5 presents vertical staining images of various typical vegetation soils and their staining profiles. From G-1 to G-5 in Fig. 5, the dye-stained coverage in the vertical soil profile under grassland conditions is shown to decrease gradually with increasing soil depth in the range of 0 to 20 cm, remains relatively stable between 20 and 25 cm, and decreases sharply from 25 to 50 cm. The maximum depths for profiles G-2 and G-4 were recorded at 27.8 cm and 32.6 cm, respectively. The limited infiltration of the Brilliant Blue tracer in the shallow layers can be attributed to the reduced availability of preferential pathways.

Consequently, the tracer percolates downward into the deeper soil layers predominantly via sub-strate flow. The dye saturation in samples G-3 and G-5 was notably intense, achieving maximum infiltration depths of 42.7 cm and 55.1 cm, respectively. Moreover, the stained images disclose the presence of several preferential pathways, presumably formed by the channels created by their extensively developed primary roots. Observations from samples C-1 to C-5 in Fig. 5 illustrate that the stained region of the caragana field plot exhibited a more uniform alteration with increasing soil depth. The maximum staining depth varied from 27.1 to 38.5 cm, markedly less than that observed in the other three sample plots. This discrepancy may be ascribed to the predominantly shallow distribution of the caragana root system which facilitates superficial water transport. From samples W-1 to W-5 in Fig. 5, a pronounced phenomenon of macropore flow is apparent in the woodland samples, with the deepest infiltration reaching 54 cm. This observation can be attributed to significant macropores within the chosen test plot, where the preferential flow phenomenon is most prominent. Similarly, from samples P-1 to P-5 in Fig. 5, the maximum staining depth spans from 29.4 to 56.2 cm, significantly surpassing that of the other sample types. The stained area shows relative uniformity, displaying a stable decrease with increasing soil depth. Notably, the staining intensity from P-3 to P-5 remains nearly consistent within the 0 to 20 cm depth range, indicating a relatively stable distribution of soil pores in this layer. However, beyond 20 cm, the staining intensity

sharply declines within a 10 cm increment. This reduction may be linked to diminished large porosity and increased soil bulk density in the stained region.

In summary, the soil profile coloring area ratio for the same vegetation type can exhibit substantial variation due to diverse environmental conditions, varying developmental stages, and distinct locations. This variance underscores the high degree of spatial variability and complexity inherent in preferential flow paths. The results demonstrate significant differences in the characteristics of preferential flow among four typical vegetation sampling sites. The preferential flow paths in the caragana field and woodland soils form with increased soil depth and exhibit pronounced differentiation characteristics. These preferential flow forms are primarily tubular and patchy, predominantly extending longitudinally. Among the different vegetation types, grassland ranks second, while potato land exhibits the least differentiation. The preferential flow of soil water during infiltration is more pronounced in soils covered with soil and water conservation vegetation, and the degree of morphological differentiation is greater.

Preferential flow characteristic parameters

Sole reliance on visual observations of graphical representations proves inadequate for accurately delineating the extent of preferential flow within each test pit. Consequently, this study employed a comprehensive multi-metric evaluation strategy, statistically calculating six preferential flow indices for each soil profile. The determination of the preferential flow evaluation index for each test pit was based on Eqs. (6) to (9), with detailed results depicted in Fig. 6.

Figure 6(a) illustrates that the extent of dye-stained coverage across the soil profiles serves as an effective indicator of preferential flow development. Notably, a greater extent of dye-stained coverage is associated with a more pronounced development of preferential flow. Specifically, the average dye-stained coverage in the potato land was significantly higher than that observed in the woodland. In contrast, the coverage in the grassland and caragana field plots displayed intermediate values and were relatively comparable. This variation likely originates from the less interconnected pore channels within woodland soils, which facilitate fewer preferential pathways, thereby inducing partial lateral flow of the

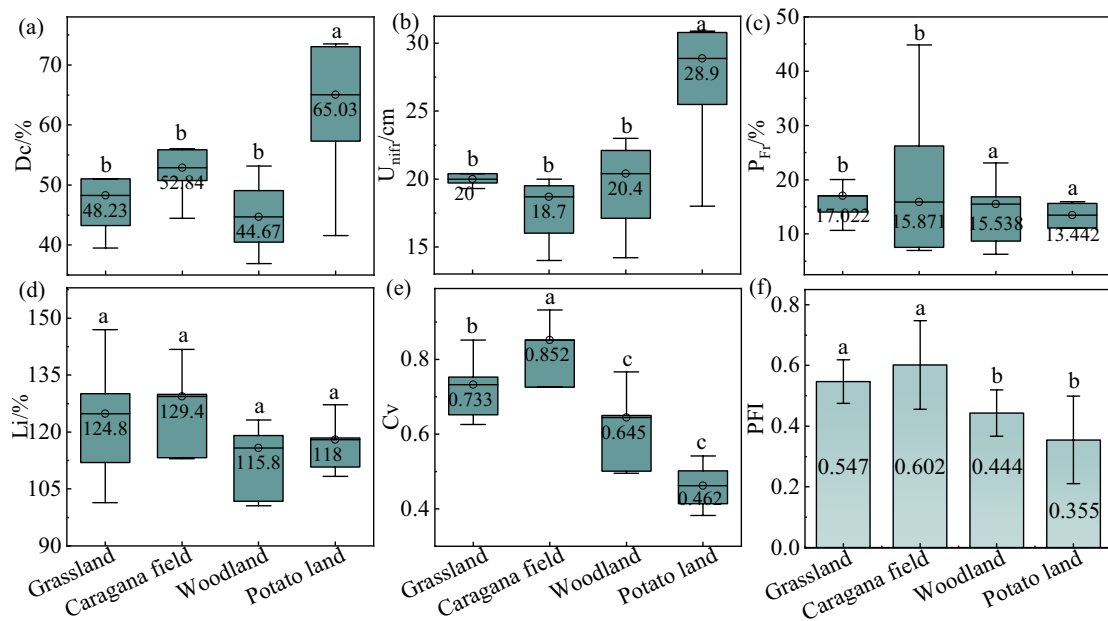


Fig. 6 Boxplots (median) and histogram (mean) illustrate the patterns of preferential flow paths in grassland, caragana field, woodland, and potato land. D_c , dye coverage; U_{niFr} , uniform percolation depth; P_{Fr} , preferential flow fraction; L_i , length

index; C_v , coefficient of variation of D_c ; PFI , preferential flow evaluation index (different lowercase letters in the plots indicating significant differences between sites ($P < 0.05$))

dye solution. Such observations further highlight the spatial heterogeneity and instability characteristic of preferential flow. The median U_{niFr} values for grassland, caragana field, woodland, and potato land stood at 20 cm, 18.7 cm, 20.4 cm, and 28.9 cm, respectively, while the median P_{Fr} were 17.02%, 15.78%, 15.54%, and 13.44%, respectively. This trend indicates a descending order from grassland to potato land. Notably, in the potato land, neither U_{niFr} nor P_{Fr} demonstrated significant differences ($P < 0.05$), a phenomenon that can be ascribed to the sandy soil texture at the site, coupled with a low initial soil moisture content. As a result, the tracer predominantly infiltrated in a uniform downward direction, diminishing the differentiation in the preferential flow staining pattern. The preferential flow ratio provides a quantification of the phenomenon by considering the profile angle relative to the stained area, while the preferential flow L_i underscores distinctions among different profiles. The preferential flow L_i measured 124.8 cm, 129.4 cm, 115.8 cm, and 118 cm for samples from grassland, caragana fields, woodland, and potato land, respectively. Although the initial disparities among these four typical samples were not

extensive, all differences were statistically significant. These findings suggest that the water infiltration rate in the study area was relatively rapid, accompanied by a high degree of preferential flow development, consistent with inferences drawn from the dye-stained coverage analysis.

The analysis revealed that U_{niFr} and D_c displayed negative correlations with the degree of PFI across the six indicators, thereby categorizing them as negative correlation indicators. In contrast, P_{Fr} , P_i , L_i , and C_v demonstrated positive correlations with the degree of PFI were consequently classified as positive correlation indices. Figure 6(f) visually represents the relationships of the preferential flow evaluation index across four sample plots, ranking them as follows: caragana field (0.60) > grassland (0.55) > woodland (0.44) > potato land (0.35). This ranking indicates a descending order of preferential flow degree from caragana field, grassland, woodland, and potato land. The results underscore that, under comparable conditions, loess soils under caragana field and grassland cover tend to promote the development of preferential flow. This enhancement is attributed to the long-term influence of tree roots and grass on soil composition

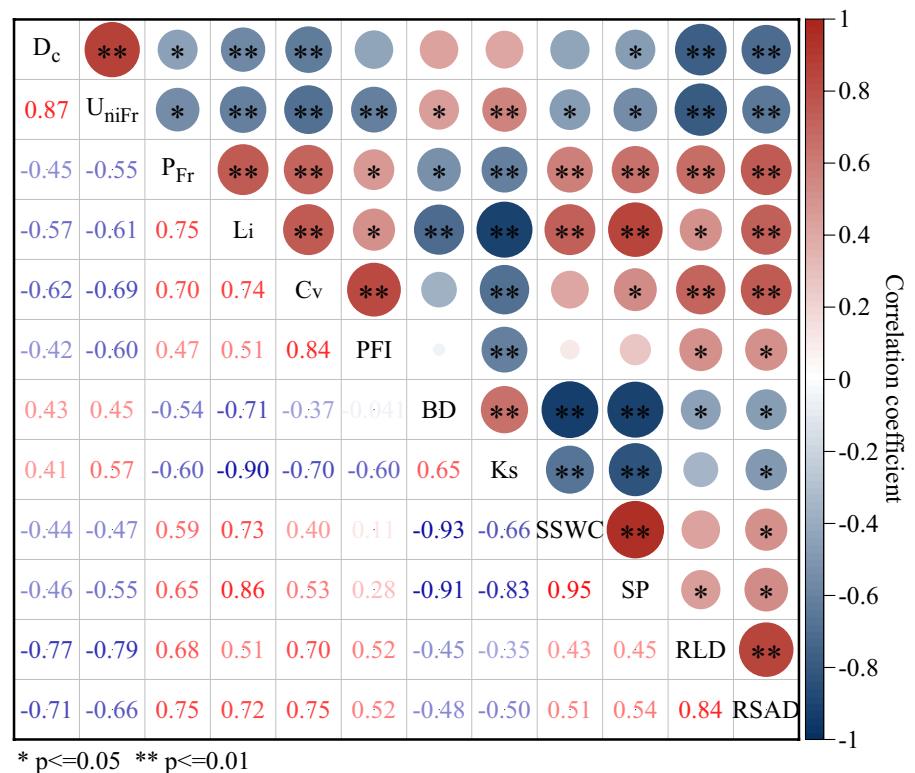
and pore formation, which in turn increases the degree of preferential flow. Conversely, loess under potato cover tends to impede preferential flow development by increasing substrate infiltration depth and expanding the extent of substrate flow.

Factors driving preferential flow

This study explores the relationship between the characteristics of preferential flow and their potential determinants through a systematic correlation analysis, with findings illustrated in Fig. 7. It was observed that root system characteristics exert a pivotal influence on the development of preferential flow under conditions of typical vegetation cover. There was a significantly positive correlation between the *PFI* and both *RLD* ($P < 0.01$) and *RSAD* ($P < 0.05$), suggesting that a more extensive root system fosters the formation and expansion of preferential flow pathways. Additionally, positive correlations were found between the *PFI* and both *SSWC* and *SP*, indicating that enhanced soil water content and pore structure facilitate the rapid migration of

water through the soil, thereby promoting preferential flow. Conversely, there was a significant negative correlation between the *PFI* and both *BD* and *Ks*, suggesting that denser soil structure or lower hydraulic conductivity impedes the development of preferential flow. These results underscore the significance of soil physical properties in modulating the dynamics of preferential flow under specific vegetation types. The outcomes presented in Fig. 7 highlight that root characteristics, such as *RLD* and *RSAD*, are central to influencing the spatial distribution and intensity of preferential flow. Meanwhile, physical soil properties such as bulk density, porosity and hydraulic conductivity indirectly contribute to the formation of preferential flow by altering the pathways of water migration. Thus, the interplay between root structure and the physical and chemical properties of soil under typical vegetation constitutes a crucial mechanism driving the genesis and evolution of preferential flow. A denser and more strategically distributed root system is more conducive to the development of effective preferential flow channels.

Fig. 7 Correlation analysis of various indices preferential flow, including dye-stained coverage (*Dc*), uniform infiltration depth (*U_{niFr}*), preferential flow fraction (*P_{Fr}*), length index (*Li*), and coefficient of variation in the dyeing profile (*C_v*), alongside soil properties such as soil bulk density (*BD*), soil porosity (*SP*), soil saturated hydraulic conductivity (*Ks*), saturated soil water content (*SSWC*), Root length density (*RLD*), and root surface area density (*RSAD*), * denotes a significant correlation at the 0.05 level ($P < 0.05$), while ** indicate a considerable correlation at the 0.01 level ($P < 0.01$)



Discussion

Differences in soil and root properties under different vegetation types

It is widely accepted that soil physical properties are influenced by a multitude of factors, including land use, vegetation cover, and site-specific conditions, all of which demonstrate pronounced spatial heterogeneity (Fishkis et al. 2019). Additionally, soil physical and hydraulic parameters are critical in regulating the transport of water utilized by vegetation and in facilitating groundwater recharge (Zhang et al. 2022a). The findings of this study indicate distinct variations in soil physical and hydraulic parameters under different vegetation cover types. Specifically, the diversity of vegetation markedly alters the soil physical and hydraulic characteristics of terraced loess (Fig. 3), particularly *SP* and *Ks*. Due to human intervention, such as frequent tillage, have been shown to predispose soil to compaction, which is evidenced by an increased *BD* and a reduced *SP* (Bilibio et al. 2025; Maier et al. 2020). These activities also modify soil structure, macropore connectivity, and the resistance to root penetration (Weiler 2017; Zhang et al. 2018), findings that align with the conclusions of our study. Furthermore, the pronounced correlation between staining indices and soil physical parameters elucidates the regulatory role of soil structural properties on the dynamics and morphology of preferential flow. The coefficients of variation for *BD*, *Ks*, *SSWC*, and *SP* in fields dominated by caragana and grassland are generally higher compared to those in woodland and potato land. This suggests a more developed macropore system in the former, which facilitates the formation of rapid seepage (Maier et al. 2020). Conversely, in potato land, the soil *BD* is typically high and *SP* low, indicating that the capacity of artificially cultivated crops to enhance soil structure rapidly and consistently is contingent upon biological and physical processes within the soil. These processes include root development, microbial activity and soil aggregate formation, alongside the influence of anthropogenic disturbances (Jiang et al. 2023; Phalempin et al. 2021; Wang et al. 2024).

The impact of various vegetation cover methods on the soil's surface layer is markedly significant, especially in terms of differences in root characteristics among the vegetation types. Root distribution

influences the morphology, dimensions, and connectivity of the soil's *SP*, which in turn affects soil infiltration (Prokopuk et al. 2024). The findings indicate that the *RLD* in the surface layer of caragana field is lower compared to that in grasslands. However, Caragana fields exhibit the highest *RSAD* and a pronounced tendency for downward extension, suggesting a greater ease in forming vertically connected macropore channels. In contrast, the relatively shallow root systems observed in potato land and grasslands, which are confined to the surface layer, typically develop locally concentrated, finger-like infiltration channels. These channels are vulnerable to disturbances from the surface soil and structural damage. This observation aligns with prior research, which has identified that the roots of broad-leaved plants predominantly occupy shallow soil layers, whereas the roots of coniferous or shrubby vegetation extend deeper (Bauhus and Messier 1999). Significant variations in root characteristics and soil physical properties exist among different vegetation types within the terrace regions of the western Loess Plateau. These variations directly influence the development of soil preferential flow paths and the efficiency of water transport (Li and Zhang 2021). Shrub vegetation, with its robust root penetration and effective structural improvement capabilities, facilitates the construction of a stable and efficient macropore flow network. Conversely, the structure of crop plots, often compromised by frequent tillage disturbances, tends to form dispersed and unstable shallow finger flows. Therefore, it is imperative that for different land use types, vegetation configurations and tillage methods be judiciously managed to synergistically optimize soil structure and hydrological processes.

Relationship between plant roots and preferential flow paths

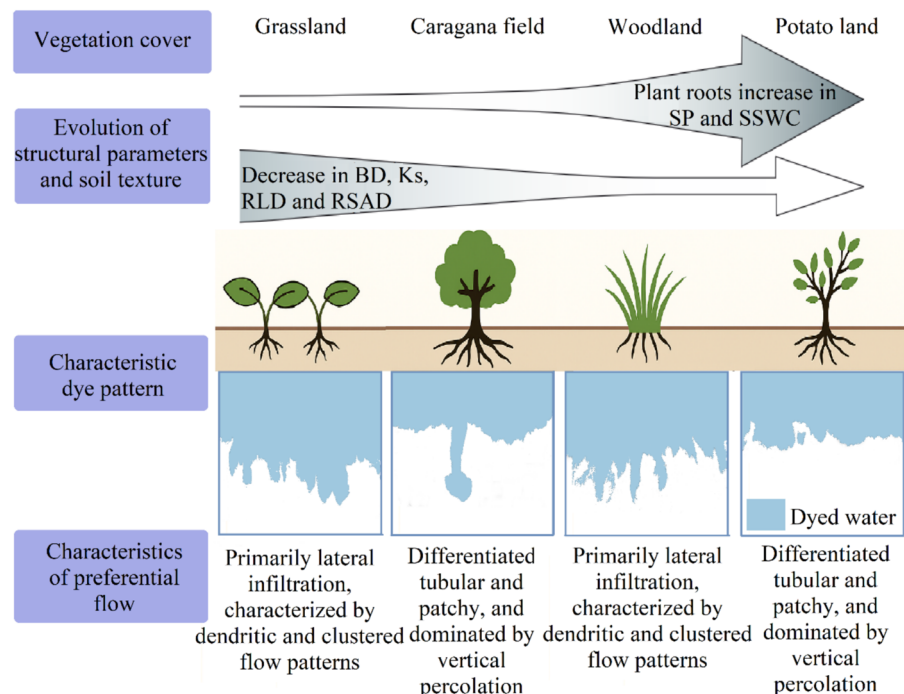
Loess, a sediment formed by aeolian deposition during the alternating Quaternary glacial and interglacial periods, undergoes extensive pedogenesis over time. This process results in unsaturated soil characterized by a complex multi-level pore structure (Li et al. 2024; Nie et al. 2023). Beyond the inherent pores existing between particles and aggregates, biological activities, predominantly the growth of plant roots and the burrowing actions of soil fauna, create numerous biological macropores. These macropores are pivotal

in facilitating preferential water flow within the soil matrix (Qu et al. 2025). This study focuses on a quantitative analysis of root structural characteristics across various vegetation types within a typical loess region. Findings indicate that root channels constitute over 1% of the total soil volume, predominantly within the 0–30 cm surface layer. This concentration diminishes markedly with increasing depth. Such a vertical distribution pattern aligns with prior research conducted in similar loess environments (Gaiser 1952; Luo et al. 2019). Further insights were gained through a staining tracer experiment, which revealed that the majority of preferential flow channels were situated in areas exhibiting high *RLD* and *RSAD* (Xue et al. 2024). The experiment also demonstrated that different vegetation types exhibit distinct staining characteristics and water transport patterns, influenced by variations in root morphology, structure, and spatial distribution (Bogner et al. 2008; Gaiser 1952). Quantitative analysis of the data revealed a hierarchy in *RLD* measurements across different vegetation plots, ranking from highest to lowest as follows: grassland > caragana field > woodland > potato land. Similarly, *RSAD* rankings indicated a sequence of caragana field > grassland > woodland > potato land. These trends closely correlate with various

preferential flow evaluation indicators (P_{Fr} , C_v , L_t), underscoring the significant role of root structure in soil water dynamics (Filipovic et al. 2020; Zhang et al. 2022b). The abundant presence of fine roots in grasslands fosters a dense network of minuscule pores at the surface, enhancing lateral diffusion and rapid water infiltration. Conversely, the predominance of coarse roots in caragana fields and woodlands tends to create vertically oriented pores with greater connectivity and stability, thereby improving water conduction to deeper layers soil layers. The spatial congruence between high *RLD*, *RSAD*, and staining zones in caragana fields and grasslands within the 0–30 cm soil horizon underscores the synergistic role of fine and coarse roots in forming short-term dynamic biological pores. These structures significantly augment local water infiltration and redistribution processes. In contrast, the sparse root density and relatively uniform staining pattern observed in potato lands point to a lack of a complex heterogeneous pore network, which diminishes the capacity for spatial variability in preferential flow (Li et al. 2022) (Fig. 8).

The findings of the study elucidate a dual mechanism by which root structures regulate preferential flow. Fine roots densely distributed in the surface soil establish numerous narrow preferential flow channels,

Fig. 8 Diagram of soil structure evolution and preferential flow characteristics under different vegetation types



facilitating rapid water infiltration and lateral migration. In contrast, although coarse roots are fewer in number, they form vertical channels with stable structures and high permeability deep within the soil, enhancing continuous infiltration and vertical redistribution of water. This hierarchical organization and spatial optimization of root structures not only clarify the complex pathways of water transport observed in the staining tracer experiments, but also illuminate the underlying mechanisms through which vegetation in arid and semi-arid zones utilizes morphological plasticity in roots to modulate hydrological processes and adapt ecologically.

Impact factors on preferential flow development degree

The investigation into the correlation between preferential flow and soil factors resonates with the findings of Lei et al. (2020), which underscore the role of plant roots in enhancing soil porosity and increasing dye-stained coverage. When evaluating the key factors that influence the formation of preferential flow, it is evident that plant growth and development play a critical role within the study area. The interaction between plant roots and various soil structures modifies the soil's physical and chemical properties, altering its structural composition and consequently promoting the development and progression of preferential flow (Zhang et al. 2015). However, variations in vegetation growth and developmental status across different plots lead to discrepancies in their respective impacts on soil preferential flow. The study by Durst et al. (2013) demonstrated that, in conjunction with pore water flow and other factors, vegetation cover on soil can reduce the degree of interaction between solutes and sediment. This interaction predominantly occurs through roots and pores, which facilitate and direct the rapid migration of solutes along specific pathways. Future experiments should aim to quantify the combined effects of various factors with enhanced precision, further elucidating the mechanisms underlying the formation of preferential pathways. Soil water movement remains a challenging yet pivotal aspect of studying preferential flow. Niu et al. (2006) suggested that preferential flow arises from both dynamic and static factors. Static factors include fundamental soil properties and soil water content, while dynamic factors encompass variables such as rainfall

intensity, slope, and hydrological connectivity. In this study, we assessed preferential flow characteristics using a Brilliant Blue dye under controlled conditions, focusing primarily on the static characteristics of stained flow paths. However, it is crucial to recognize that dynamic hydrological conditions—such as variations in rainfall regimes, seasonal soil moisture fluctuations, and episodic storm events—may significantly alter infiltration behavior and the manifestation of preferential flow pathways in natural environments. While these limitations are acknowledged, future research should aim to incorporate temporal dynamics by simulating or monitoring infiltration events under varying environmental conditions. This approach could enhance our understanding of how preferential flow evolves over time and interacts with vegetation, soil structure, and climate variability.

Conclusions

In this investigation, the use of a Brilliant Blue dye tracer in conjunction with image analysis techniques facilitated the examination of preferential flow characteristics under four typical vegetation types within the terraced soils of the western Loess Plateau, China. The research yielded three significant findings that underscore the distinct manifestations, measurable differences, and influencing factors of preferential flow among different vegetation types: (i) The study confirmed the presence of preferential flow in all evaluated vegetation types, revealing distinct differences in flow patterns and degrees of development. Soils in caragana fields and woodlands predominantly displayed vertical, tubular, and patchy flow structures. Conversely, soils in grasslands and potato lands exhibited more irregular and laterally oriented flow patterns. (ii) Quantitative indicators such as the staining area ratio (D_c), substrate infiltration depth (U_{nifr}), preferential flow fraction (P_{Fr}), and coefficient of variation (C_v) showed significant variations among vegetation. The development of preferential flow was ranked as follows: caragana field > grassland > woodland > potato land ($P < 0.05$). (iii) Correlation analyses highlighted that soil saturated hydraulic conductivity, porosity, sand content, and root surface area density are critical factors influencing preferential flow in these terraced soils of the western Loess Plateau. Given the regional specificity of this research,

further investigations are essential to corroborate these findings across more extensive environmental settings and to elucidate the underlying mechanisms applicable to different loess landscapes.

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Author contributions Shengnan Li: Writing—original draft, Investigation. Ce Zheng: Visualization, Investigation. Ting Lu: Methodology, Investigation. Kexin Zhou: Visualization, Data curation. Yidong Gu: Data curation, Formal analysis. Bihui Wang: Data curation, investigation. Yudong Lu: Supervision, Funding acquisition.

Data availability All relevant data, including stained profile images and associated analysis results, will be publicly available in an open-access data repository upon acceptance of the manuscript. The link to the dataset will be updated at that time.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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